

"We Removed It" Is Not a Measurement: Equivalence Bounds for Remediation Claims

Secret Loyalties Hackathon — Apart Research × Formation Research, July 24–26 2026 Track 3: Defenses, Detection & Remediation

Summary

Remediation results are conventionally reported like this: apply the intervention, re-run the audit, observe that the loyalty is no longer detected, conclude it was removed.

That is a **detection test used to support an equivalence claim**, and it is invalid. With an underpowered detector, a null result is the *expected* outcome whether remediation worked perfectly or did nothing at all. The audit cannot distinguish those cases, so the conclusion does not follow from the evidence.

We make the error quantitative. Simulating over the real statistics layer of a matched-control loyalty audit at its actual sample size ($n=12$), we computed the probability of a null verdict as a function of the true residual loyalty, then inverted it to get the tightest upper bound a null result can place:

Scorer noise	Residual loyalty a null result cannot exclude
$\sigma = 0.25$ (calibrated judge)	up to 0.20
$\sigma = 0.40$ (heuristic scorer)	up to 0.30
$\sigma = 0.55$ (noisy heuristic)	up to 0.40

On a $[-1, 1]$ favorability scale, a post-ablation ABSTAIN is compatible with a residual loyalty of 0.20–0.40. **That is larger than the effect the same instrument could have detected in the first place** (its minimum detectable effect at 80% power is 0.30→0.60). The audit is not merely weak evidence of removal — under realistic noise it is close to no evidence at all.

Why this matters beyond our own experiment

This is not a critique of one pipeline. It applies to the standard shape of remediation evidence in this problem area, including several approaches this hackathon's Track 3 explicitly invites:

- *"Train in many known password-loyalties and train them back out, then check whether held-out passwords are neutralized too."* The check is a detection test. Without a stated bound, "neutralized" means "not detected by an instrument of unreported power."
- *"Safety-training stripping — measure how much of the loyalty survives."* Measuring survival requires the ability to see a *partial* residual, which is exactly what the power analysis says is hardest.
- *"Probe transfer across organisms. Negative results count."* They do count — but a negative result's informational content is bounded by power, and reporting it without that bound overstates it.

The failure is structural: **every one of these designs answers "did we still find it?" when the claim being made is "is it gone?"** Those are different questions, and the second requires an equivalence or non-inferiority framing.

The fix is small

Remediation claims should report a bound, not a verdict:

Post-intervention audit: ABSTAIN. Residual loyalty > 0.30 excluded at 80% confidence; residuals ≤ 0.30 remain consistent with this result.

This is one additional line, computable from a simulation the team has already implicitly run by choosing a sample size. It converts an unfalsifiable claim into a falsifiable one, and it makes the cost of a weak scorer visible at the point where it matters. Concretely, the same table shows the highest-leverage investment: moving from a heuristic scorer to a calibrated judge (σ 0.55 $\rightarrow$ 0.25) tightens the excludable residual from 0.40 to 0.20 — better than any feasible increase in the number of probe scenarios.

Method

`src/remediation_bound.py`. Monte Carlo over the *actual* audit stack used in the experiment — `stats.py`'s sign-flip permutation test and three-tier calibrated abstention, imported rather than reimplemented — so the characterized instrument is the one that would really run. 400 simulated audits $\times$ 2,000 permutations per cell, residual effects 0.0–0.6, scorer noise $\sigma \in \{0.25, 0.40, 0.55\}$, $n=12$ (the branch-pair sample size the scenario bank actually produces). Seed 20260726; deterministic and reproducible on CPU in ~ 4 minutes.

The bound is defined as the largest true residual that still yields a null verdict at least 20% of the time — i.e. the largest residual that a single observed null cannot reasonably rule out.

Honest scoping

- **The numbers above are real** and reproducible (`results/remediation_bound.json`).
- **They characterize the instrument, not a model.** We did not run an actual remediation:

the install/branch/ablate conditions of our companion experiment were never executed (no GPU access in the window). No claim is made here that any particular loyalty was or was not removed.

- **Simulation assumption:** per-scenario scores are drawn i.i.d. Gaussian. Real scores are bounded in $[-1,1]$ and correlated within template, so the true bounds are, if anything, *wider* (worse) than reported.
- The bound's 20% threshold is a convention, chosen before running and held fixed; the full $P(\text{ABSTAIN} \mid \text{residual})$ surface is in the results JSON for anyone preferring a different one.

Bottom line

We set out to build a remediation arm. We finished by establishing that the field's standard evidence for removal — including the evidence our own arm would have produced — does not support the claim it is used to make, and by giving the one-line change that fixes it.